

# Statistical Analysis Plans

## Project so far

- + Identified a dataset, response variable, and research question
- + Data description and EDA

What's next: using tools from class to address your research question

But first: we should plan the analysis we want to do!

# Statistical Analysis Plan (SAP)

A **statistical analysis plan** (SAP) describes:

- + the research questions we are trying to answer
- + the data we will use to answer the research questions
- + how we will use statistical methods (models, diagnostics, hypothesis tests, etc.)
- + how we will present our results

**Why?**

- + having a plan ahead of time helps organize our work
- + research questions should inform analysis and methods (not the other way around)
- + helps determine which data we need to collect
- + make sure we're doing something reasonable before we do a ton of work!

# Statistical Analysis Plan (SAP)

- + What is/are the research question(s) or objectives?
- + What are the variables that will be measured in the study? For each variable, what will the data look like?
- + What is the study design, and why is the study set up this way?
- + How will data and any patterns/results be summarized? (i.e., how you will do exploratory data analysis).
- + What will be examined statistically? Link directly back to the research questions.
- + What alternative strategies should be considered, and when?
- + What are the statistical results that will be presented?

# Today

- + What will be examined statistically?
  - + Which models will you fit?
  - + How will you assess the suitability of these models? (diagnostics, etc.)
  - + How will you use the models to address the research question(s)? (hypothesis tests, confidence intervals, etc.)
  - + What software will you use? (software language, specific functions)
- + What alternative strategies should be considered, and when? (i.e., what will you do if assumptions are violated? How would you modify the model or statistical analysis?)

**Your task today:** Work through these questions (handout), and discuss your ideas with Dr. Evans